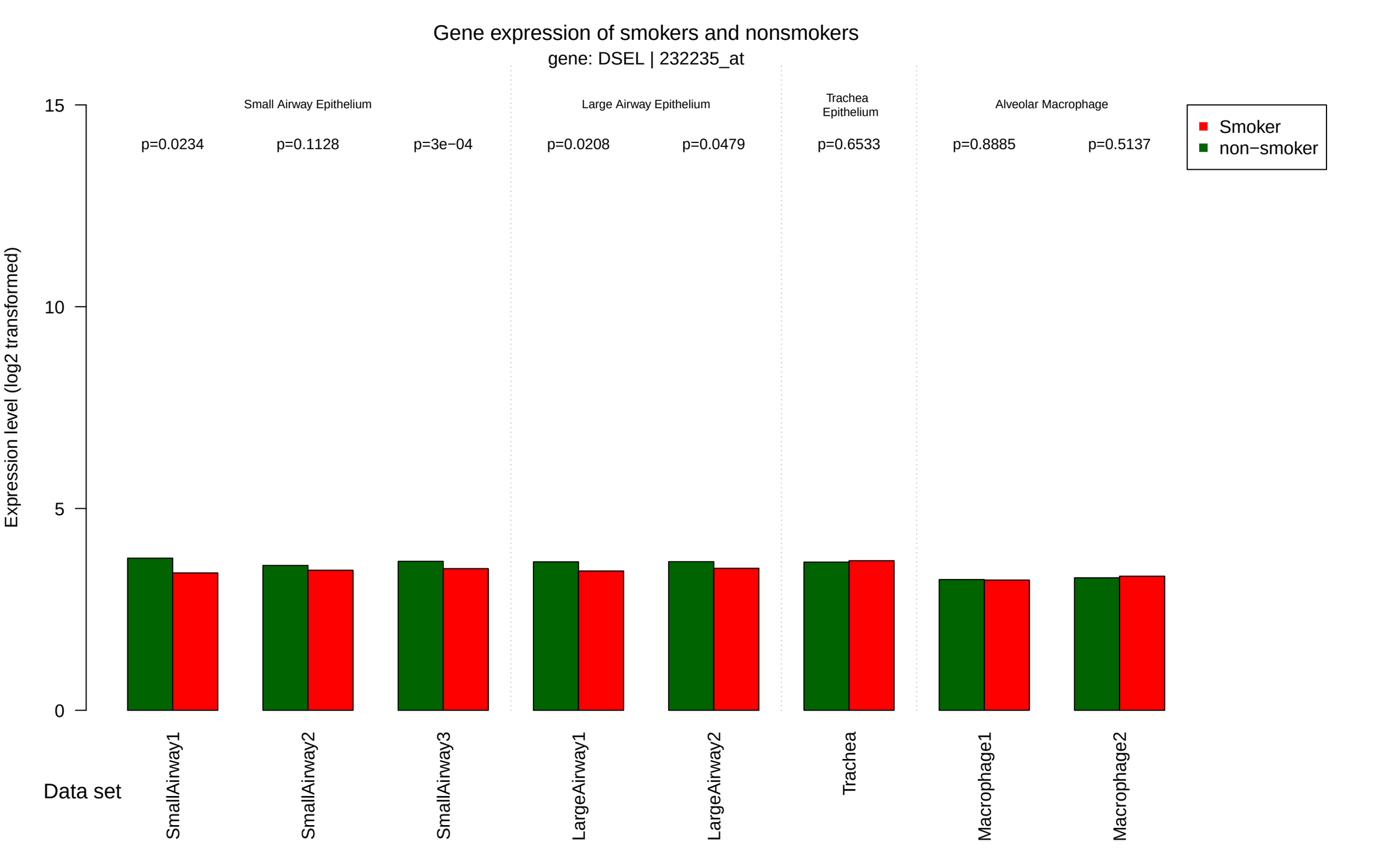
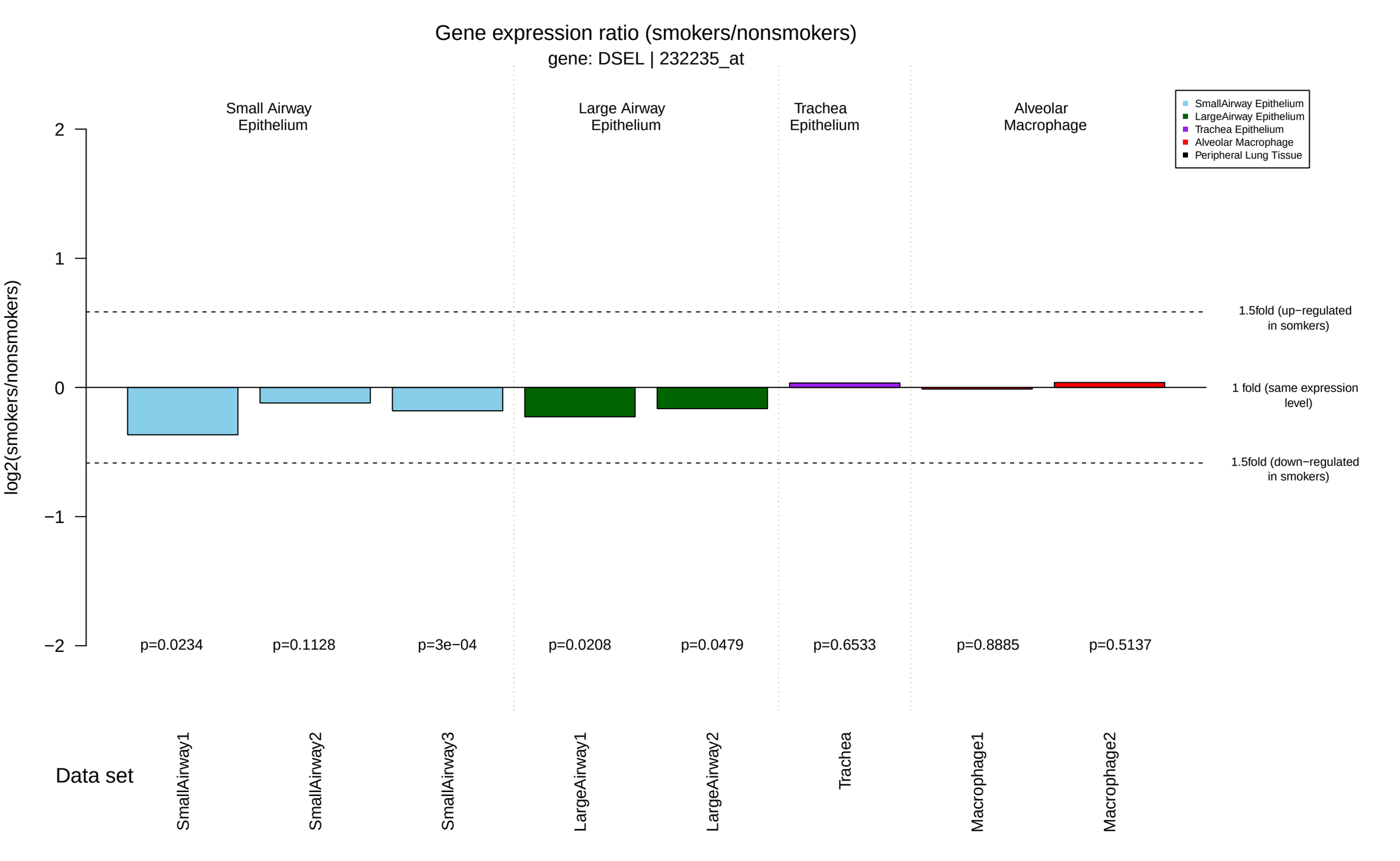
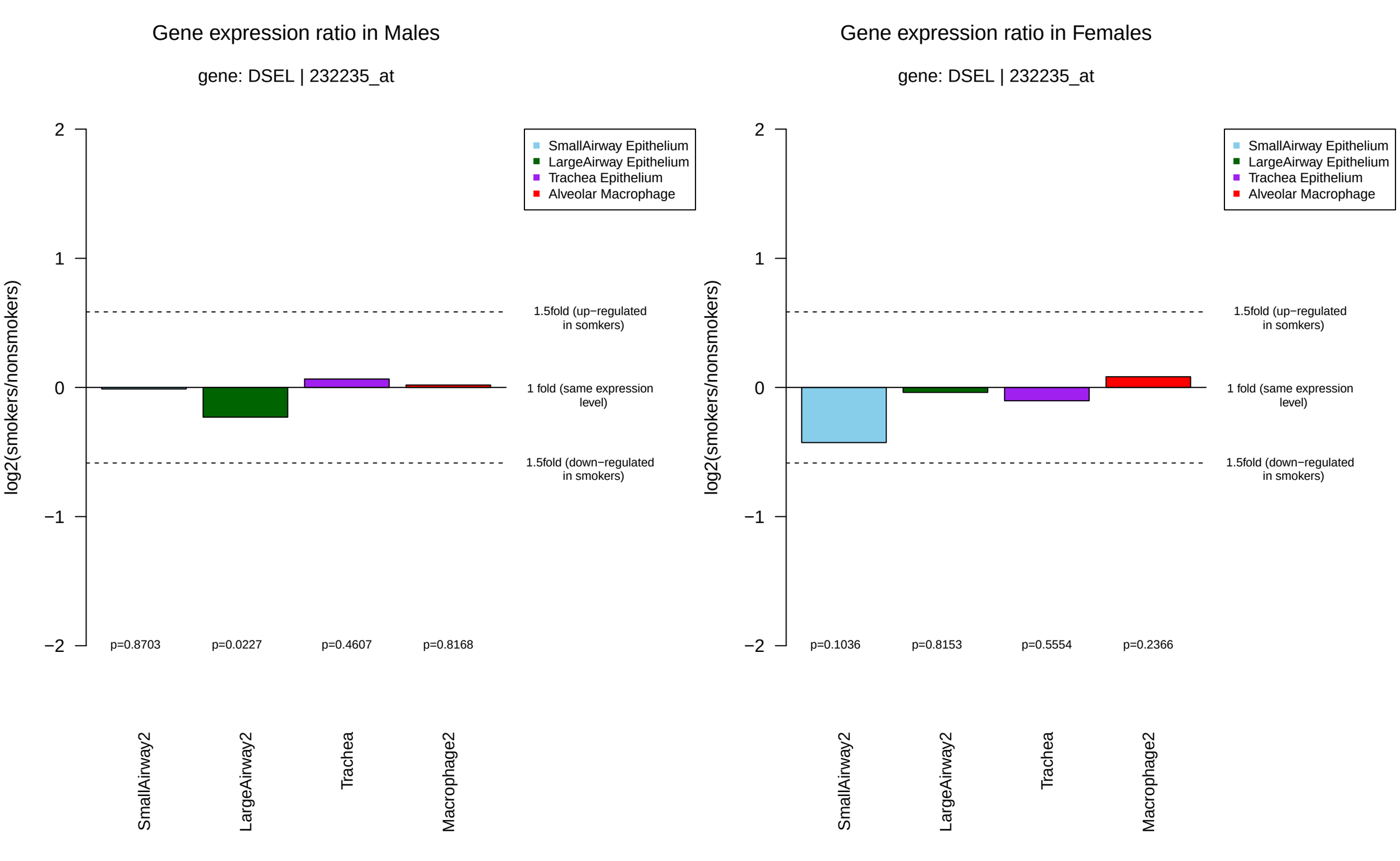
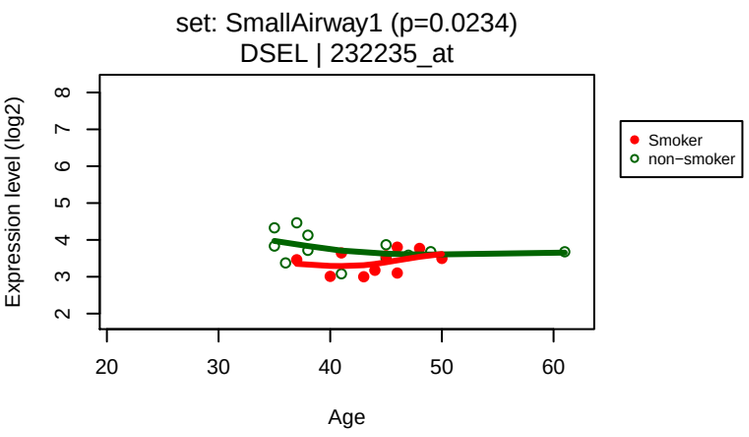
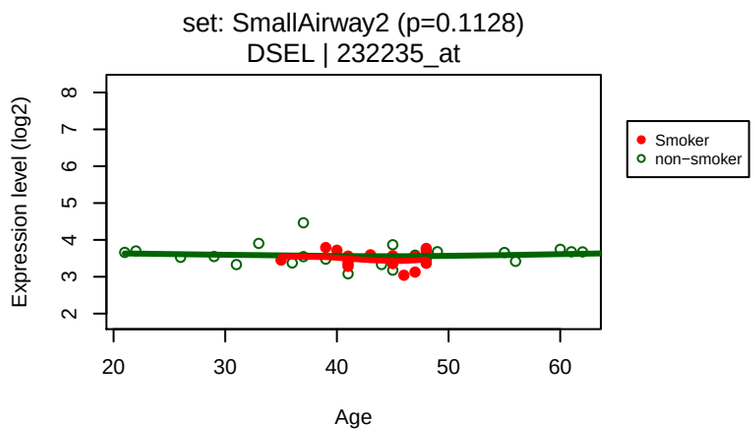
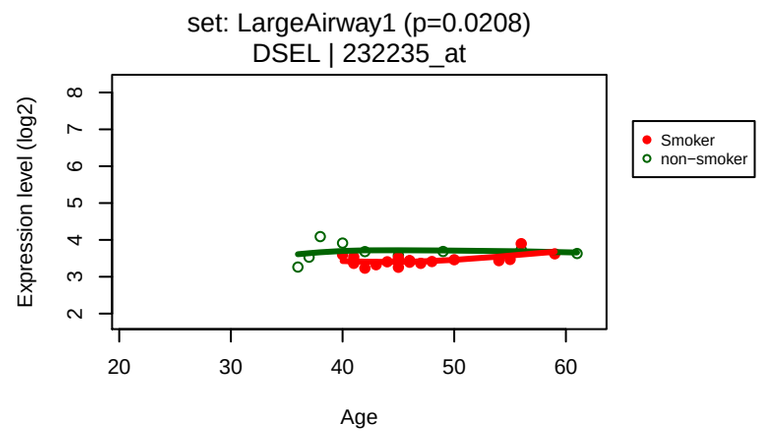

Gene expression of smokers and nonsmokers
gene: DSEL | 232235_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0234
p=0.1128
p=3e−04
p=0.0208
p=0.0479
p=0.6533
p=0.8885
p=0.5137
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

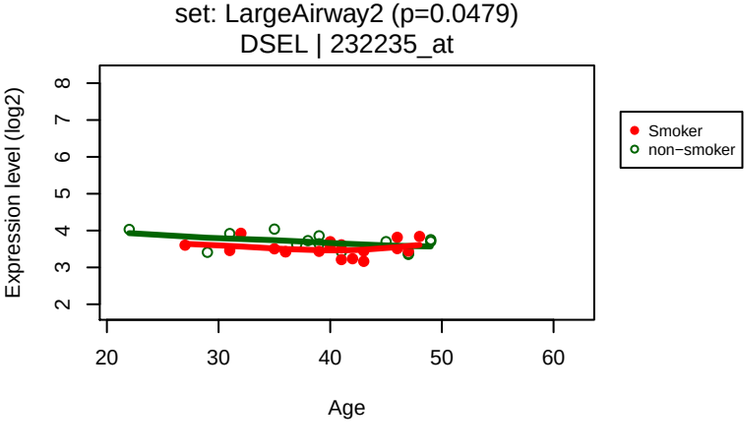

Gene expression ratio (smokers/nonsmokers)
gene: DSEL | 232235_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0234
p=0.1128
p=3e−04
p=0.0208
p=0.0479
p=0.6533
p=0.8885
p=0.5137
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

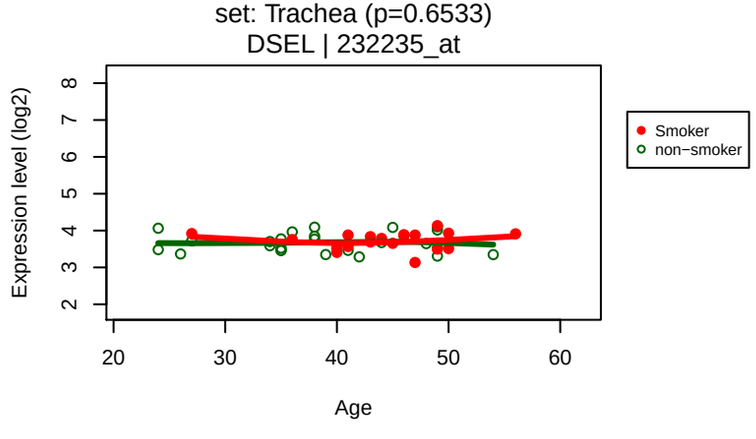

Gene expression ratio in Males
gene: DSEL | 232235_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8703
p=0.0227
p=0.4607
p=0.8168
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DSEL | 232235_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1036
p=0.8153
p=0.5554
p=0.2366
SmallAirway2
LargeAirway2
Trachea
Macrophage2

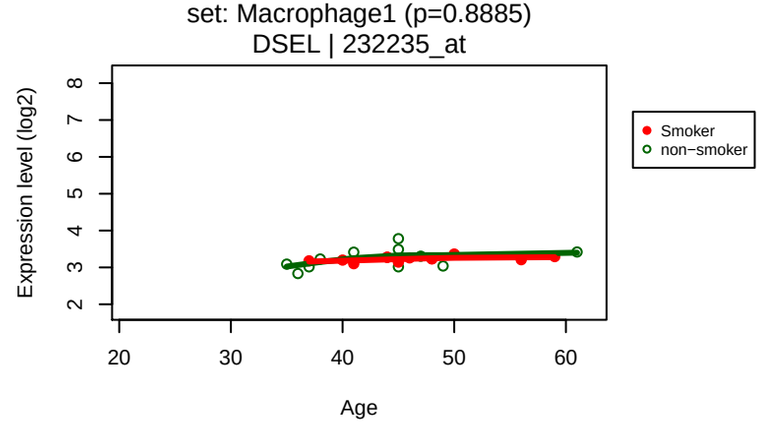

set: SmallAirway1 (p=0.0234)
DSEL | 232235_at
Expression level (log2)
Age
Smoker
non−smoker

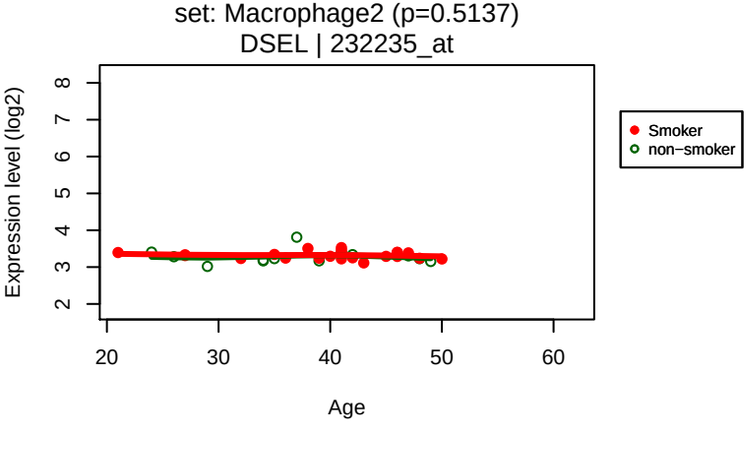

set: SmallAirway2 (p=0.1128)
DSEL | 232235_at
Expression level (log2)
Age
Smoker
non−smoker

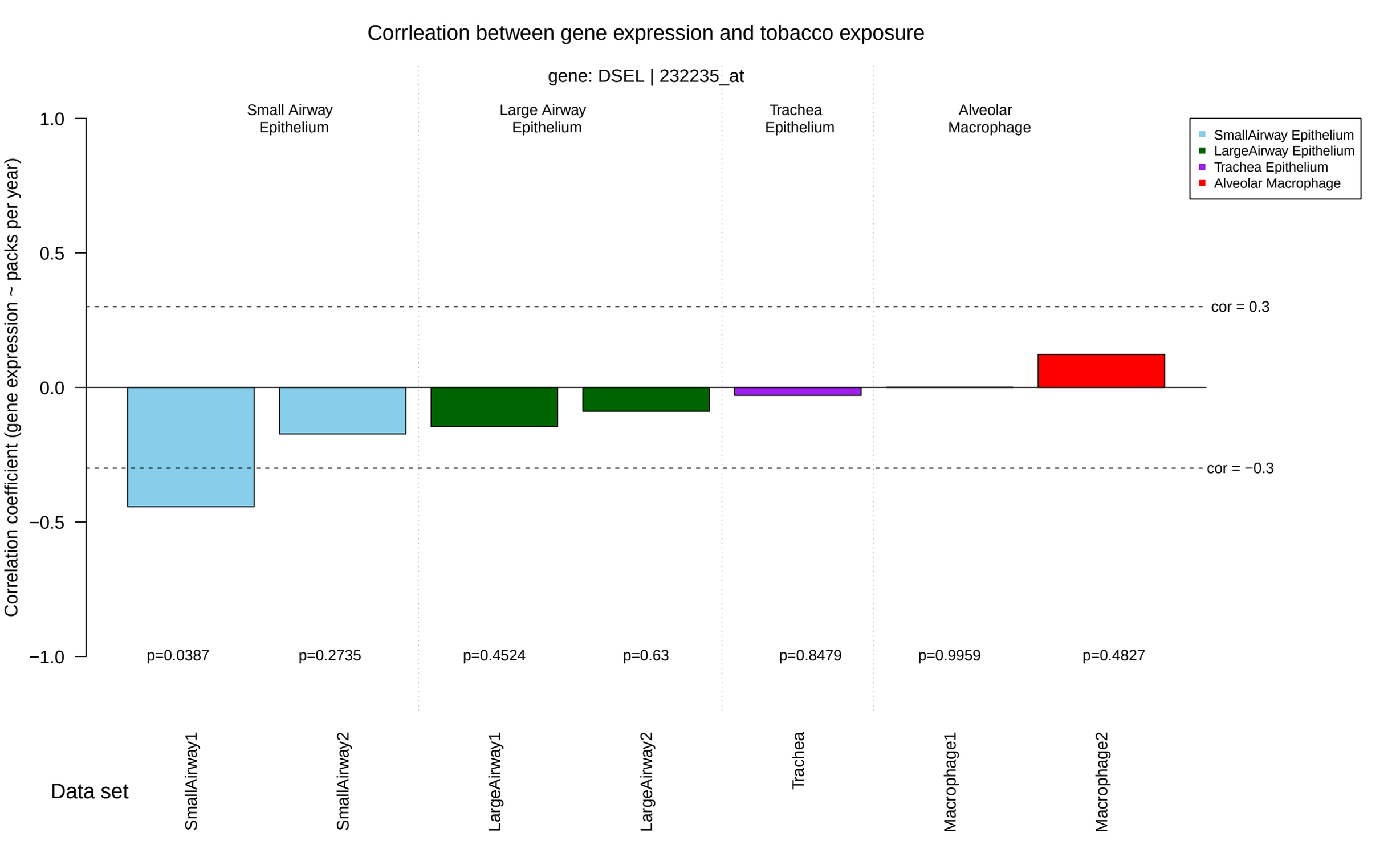

set: LargeAirway1 (p=0.0208)
DSEL | 232235_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0479)
DSEL | 232235_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6533)
DSEL | 232235_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8885)
DSEL | 232235_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5137)
DSEL | 232235_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DSEL | 232235_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0387
p=0.2735
p=0.4524
p=0.63
p=0.8479
p=0.9959
p=0.4827
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2